Q1) Use pandas' read_sql() function to load all records from a 'restaurants' table (with columns like name, cuisine, rating) in your local SQLite database into a DataFrame, then display the first 5 rows.

```
import sqlite3
import pandas as pd

conn =
sqlite3.connect("restaura
nts.db")

df = pd.read_sql("SELECT
* FROM restaurants",
conn)

print(df.head(5))

conn.close()
```

Q2) In a Jupyter Notebook, use the %sql magic command to run a SQL query that selects all movies with a rating above 8 from a 'movies' table, and display the results in the notebook.

```
%%sql
SELECT *
FROM movies
WHERE rating > 8;
```

Q3) Combine SQL and Python to analyze Zomato-style order data: use pandas read_sql() to load the last 100 orders from an 'orders' table, then use DataFrame methods to find the top 3 most ordered food items.

```
import sqlite3
import pandas as pd

conn =
sqlite3.connect("zomato.d
b")

query = """
SELECT *
FROM orders
ORDER BY order_date DESC,
order_id DESC
LIMIT 100;
"""

df = pd.read_sql(query,
conn)

item_counts =
df["food_item"].value_cou
nts()

top_3 =
item_counts.head(3)

print("Top 3 Most Ordered
Food Items:")
print(top_3)

conn.close()
```

Q4) After loading IPL match data from a 'matches' table into a pandas DataFrame, create a bar chart showing the number of matches played by each team using matplotlib or seaborn.

Hint: Use groupby() in pandas to count matches per team before plotting.

```
import sqlite3
import pandas as pd
import matplotlib.pyplot
as plt

conn =
sqlite3.connect("ipl.db")
df = pd.read_sql("SELECT
* FROM matches", conn)
conn.close()

team_matches =
pd.concat([
    df["team1"],
    df["team2"]
]).value_counts()

team_matches.plot(kind="b
ar")

plt.title("Number of
Matches Played by Each
IPL Team")
plt.xlabel("Team")
plt.ylabel("Number of
Matches")
plt.xticks(rotation=45,
ha="right")
plt.tight_layout()
plt.show()
```

Q5) Use ChatGPT or Copilot to help you write a Jupyter Notebook cell that connects to a SQLite database, runs a SQL query to fetch all users who have placed more than 5 orders, and loads the result into a pandas DataFrame.

```
import sqlite3
import pandas as pd

conn =
sqlite3.connect("zomato.d
b")

query = """
SELECT user_id, COUNT(*)
AS total_orders
FROM orders
GROUP BY user_id
HAVING COUNT(*) > 5;
"""

df = pd.read_sql(query,
conn)

display(df)

conn.close()
```